

MINGREN SHEN

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EDUCATION

University of Wisconsin-Madison

Overall GPA: 3.9 / 4.0

M.S. Computer Sciences, May 2019

M.S. Materials Science and Engineering December 2020

Ph.D. Materials Science and Engineering, December 2021

University of Chinese Academy of Sciences M.S. Physics, July 2016

Overall GPA: 3.9 / 4.0

Nanjing University B.S. Physics, July 2013

Overall GPA: 3.8 / 4.0

WORK EXPERIENCE

Software Development Engineer

Fulltime at Amazon.; April. 2022 - Present

- Built and evolved machine learning systems for demand forecasting and financial projections across Amazon Grocery, serving 500+ global sites and reducing forecasting errors by 100+ basis points
- Developed GenAI-powered diagnostic tools using Bedrock to reduce on-call resolution time from hours to 5 minutes
- Architected scalable solutions on AWS infrastructure (Glue, Lambda, Step Functions) processing large scale data

Machine Learning Engineer

Internship at Ansys Inc.; June. 2021 - August. 2021

- Built Convolution Neural Network (CNN) to correct the lower resolution 3-D flow fields to higher resolution ones with less time and few resources and achieved 40% acceleration in speed and 20% less memory usage

SKILLS

- **Programming:** Python, Java, C, HTML/CSS, JavaScript
- **Software & Tools:** Linux, Git, Bash, SQL, Pandas, Scikit-learn, OpenCV, Spark, Flask, Django, Docker, Bootstrap
- **AWS Services:** Glue, Lambda, Step Functions, Bedrock, S3, EC2, SageMaker
- **Frameworks:** Keras, PyTorch, Tensorflow
- **Areas:** Deep Learning, Machine Learning, Software Development, Computer Vision, Generative AI

SELECTED PROJECTS

ASIN-Forecasting: Demand Forecasting System for Amazon Grocery

January 2023 - January 2025

- Built an automated forecasting system to predict customer demand for Amazon Grocery and Whole Foods Market, generating daily forecasts for downstream services including purchasing and inventory management, reducing forecasting errors by 100+ basis points
- Evolved forecasting models from Bayesian inference to tree-based boosting ensemble models and transformer-based deep learning models, implementing the solution on AWS infrastructure using Glue, Lambda, and Step Functions etc.
- Developed a GenAI agent using Bedrock to accelerate on-call diagnosis, reducing resolution time from hours to 5 minutes, and built web portals enabling customers to analyze forecasting data through multiple GenAI-powered agents

Topline: Automated Forecasting System for Grocery Store Financial Reporting

April 2022 - December 2022

- Built an automated forecasting system (Topline) to generate quarterly financial report projections, predicting sales, operational volumes, and revenue for all North American Amazon Grocery stores
- Reduced financial team preparation time from weeks to 10 hours while decreasing forecasting errors by 20%
- Designed the system to support multiple business channels and store types by incorporating demographic changes, historical sales data, zip code service areas, and other key signals

Full Stack Website for Defect Detection Using YOLO Models

June. 2021 - December 2021

- Build website that helps Oak Ridge National Laboratory users to identify defects in microscopic images
- Front-End is a combination of HTML/CSS, JavaScript, and Bootstrap to support image uploading on client side
- Back-End is RESTFUL API on web server side with Flask, SQLite, Tensorflow to run deep learning models to detect defects from user input images
- Easy to use website interface and display defect detection results with interactive graphs and statistics

SELECTED PUBLICATIONS

1. **Mingren Shen** et al. *Machine learning for interpreting coherent X-ray speckle patterns*, Computational Materials Science 230 (2023): 112500.
2. **Mingren Shen** et al. *A deep learning based automatic defect analysis framework for In-situ TEM ion irradiations*, Computational Materials Science 197 (2021): 110560.
3. **Mingren Shen** et al. *Assessing Graphbased Deep Learning Models for Predicting Flash Point*, Molecular Informatics (2020), 39, 1900101.